

End-to-End Model Development Document

CMPL-REG-24

Two-stage complaint classifier · leakage audit · dual-judge panel · DPO/RLAIF

Generated 2026-08-04 03:35 UTC
Artifact: docs/complaint_model/05_end_to_end_model_development.pdf
Production gate: logistic_regression (l1, C=4.0) @ 0.491
Test ROC-AUC 0.9362 · PR-AUC 0.9904 · gap 0.0601

Abstract

This end-to-end Model Development Document records the full lifecycle of CMPL-REG-24, a two-stage consumer-complaint classifier on 4,000 curated CFPB narratives (87.5% regulatory / 12.5% non-regulatory). Stage 1 is a millisecond TF-IDF gate — production champion **logistic_regression** (l1, C=4.0, cut-off 0.491) with test ROC-AUC 0.9362, PR-AUC 0.9904, train/test gap 0.0601. Stage 2 is RAG + LLM over a 24-category regulation taxonomy (exact agreement 0.25, family 0.39 on n=101). The document also covers the 5% scoring holdout, the v1→v3 data update log, the leakage audit (split contamination + weak-label target leakage), the dual-judge panel (NIM + OpenAI as first-class judges, not fallbacks), the DPO/RLAIF post-training loop, batch ingestion, and MCP/A2A serving. Disposition: **Approve with Conditions** pending a human-adjudicated golden set.

1 · Objective, risk tier, and scope

Tier-2 two-stage classifier: stage-1 gates regulatory nexus; stage-2 assigns a 24-category label with citations. Scope covers data (v1-v3), splits, stage-1/2, leakage audit, dual-judge panel, DPO/RLAIF, batch ingestion, and MCP/A2A serving.

3 · Data update log (Table)

	version	non-reg	cut-off	ROC-AUC	gap
	v1 · generic pull only	135 (3.4%)	0.639	0.849	0.147
v2 ·	+ targeted service-issue	500 (12.5%)	0.397	0.945	0.031
v3 ·	+ cosine fuzzy dedup (cur	500 (12.5%)	0.491	0.936	0.060
	v1 → v3 net	+365 minority	—	+0.087	−0.087

v3 is the honest headline after cosine fuzzy dedup.

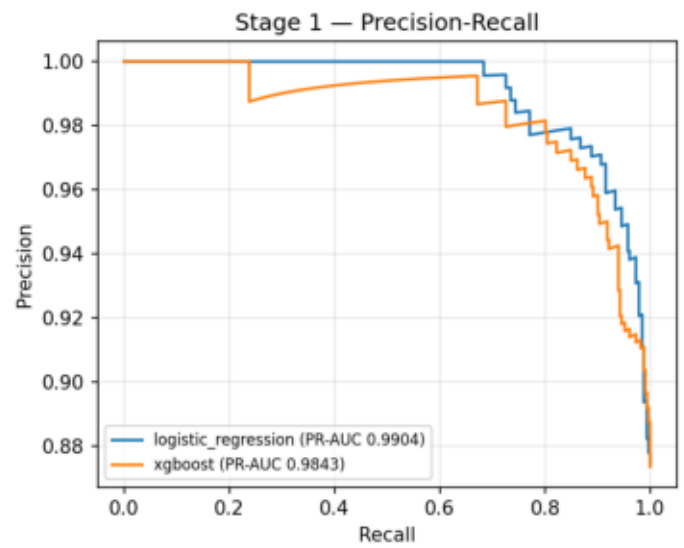
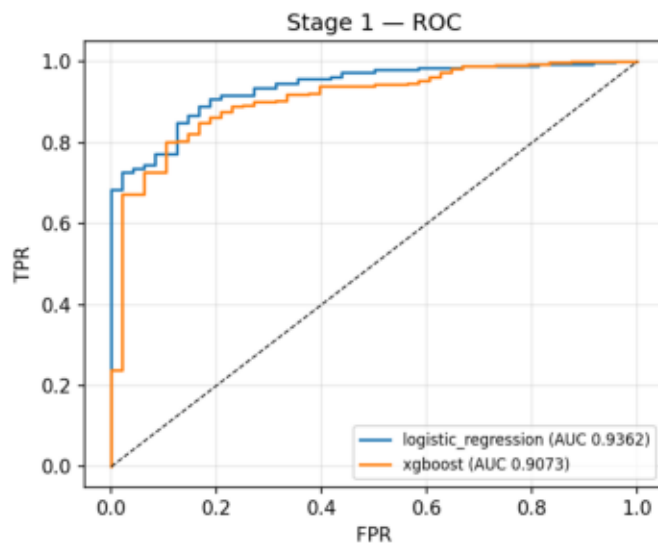
3 · Split protocol (Table)

split	n	reg	non-reg	rate
train (80%)	3,040	2,660	380	87.50%
validation (10%)	380	333	47	87.63%
test (10%)	380	332	48	87.37%
scoring holdout (5% of full)	200	175	25	87.50%

4 · Production stage-1 leaderboard (Table)

	model	params	valPR	thr	PR	ROC	gap	F1	prec	rec
logistic_regress		l1, C=4.0	0.9936	0.491	0.9904	0.9362	0.0601	0.9401	0.9592	0.9217
n_estimators=300, max_de			0.988	0.545	0.9843	0.9073	0.0927	0.9412	0.9426	0.9398

4 · Stage-1 ROC / PR curves



5 · Leakage — split contamination (Table)

check	count	expected	status
exact duplicate (normalized hash)	0	0	pass
near-duplicate (200-char prefix)	0	0	pass
near-duplicate (TF-IDF cosine ≥ 0)	2	$\leq 1\%$ of test	pass

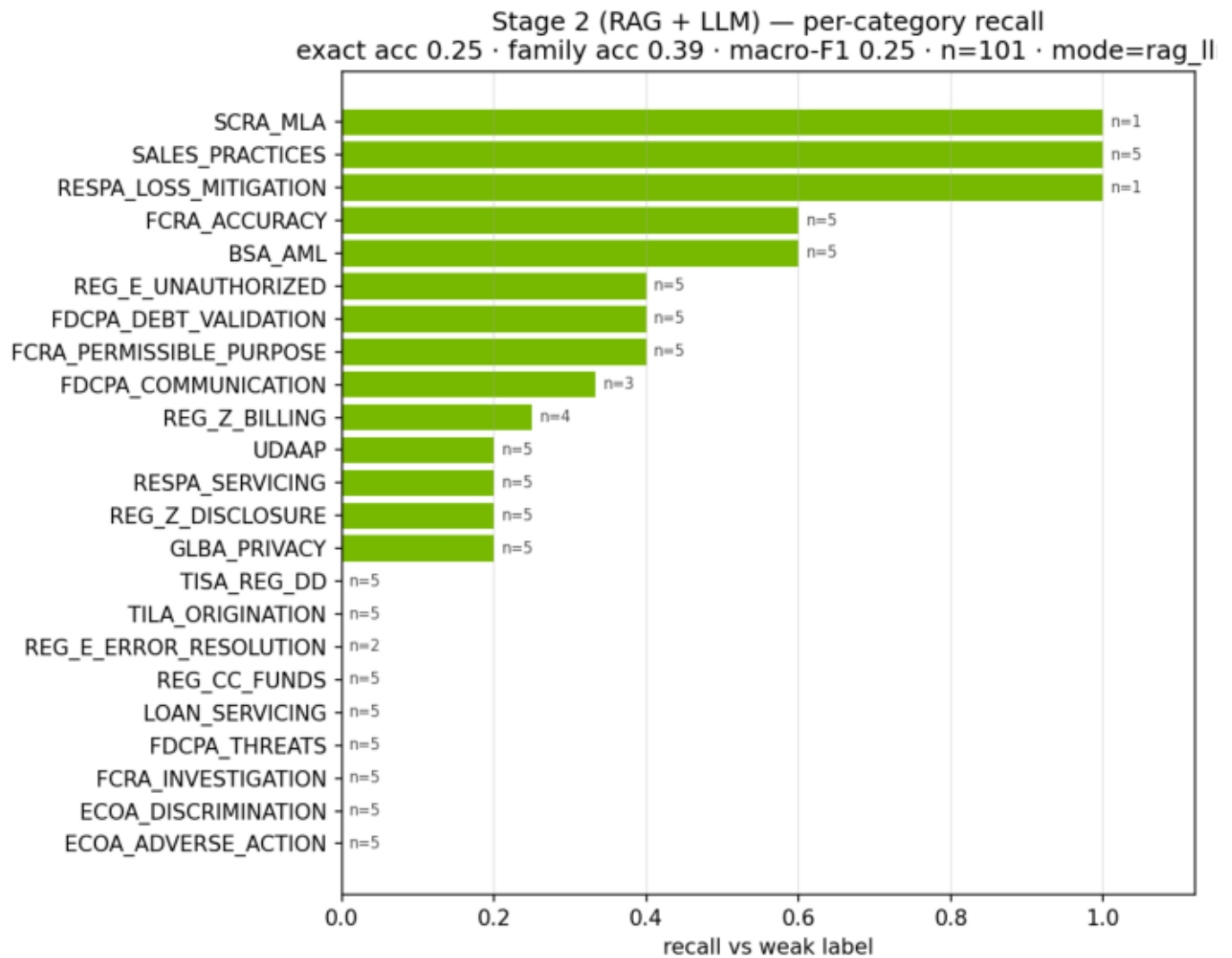
5 · Leakage — evaluation slices (Table)

slice	n	reg-rate	ROC	PR	F1min	bAcc
full test set (headline)	380	0.87	0.937	0.990	0.626	0.828
metadata-labeled (leakage)	81	0.41	0.907	0.895	0.818	0.814
narrative-regex-labeled (single class: regulatory by content)	299	—	—	—	—	—

6 · Stage-2 top per-label recall (Table)

label	support	recall
RESPA_LOSS_MITIGATION	1	1.0
SALES_PRACTICES	5	1.0
SCRA_MLA	1	1.0
BSA_AML	5	0.6
FCRA_ACCURACY	5	0.6
FCRA_PERMISSIBLE_PURPOSE	5	0.4
FDCPA_DEBT_VALIDATION	5	0.4
REG_E_UNAUTHORIZED	5	0.4

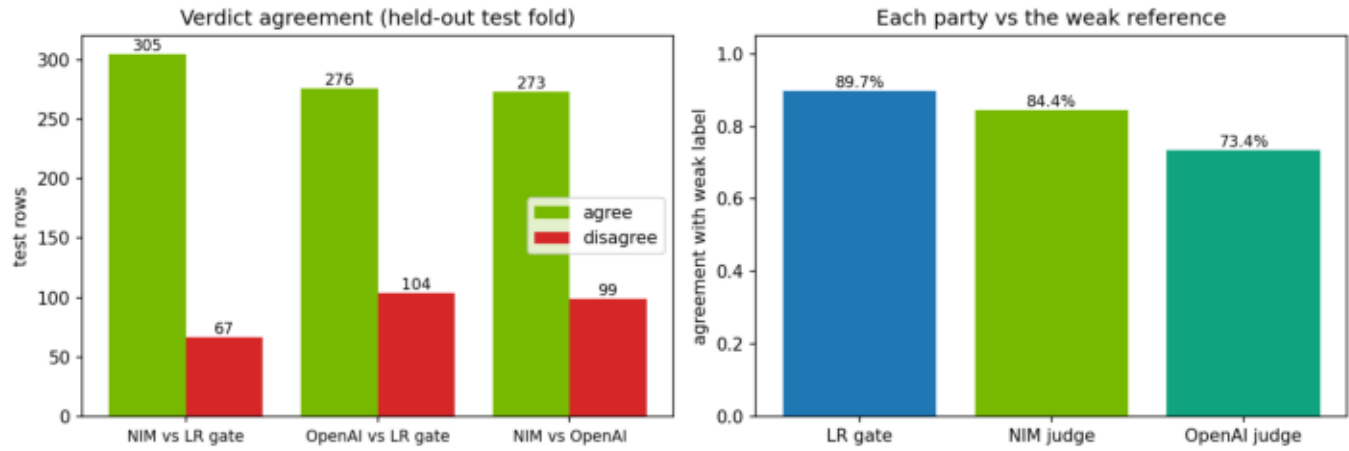
6 · Stage-2 per-label recall



7 · Dual-judge agreement (Table)

pair	n	agree	disagree	rate	κ
NIM vs LR gate	372	305	67	82.0%	0.0451
OpenAI vs LR gate	380	276	104	72.6%	0.2218
NIM vs OpenAI	372	273	99	73.4%	0.1196
LR gate vs weak label	380	341	39	89.7%	0.5833

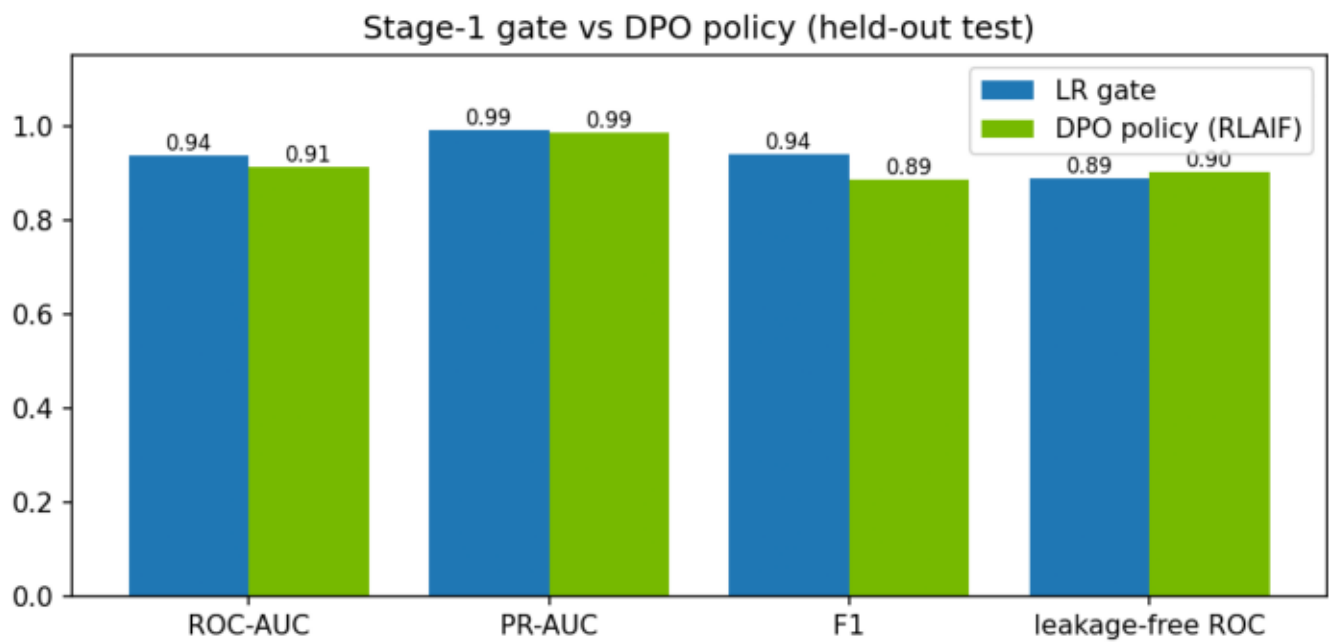
7 · Dual-judge agreement figure



8 · DPO / RLAIF vs LR gate (Table)

party	thr	ROC	PR	F1	leak-free	vs LR
LR gate (production)	0.491	0.9362	0.9904	0.9401	0.8876	1.0
DPO / RLAIF policy	0.885	0.9134	0.9866	0.8852	0.9028	0.8342

8 · DPO vs gate



13 · Disposition

Stage-1 production gate: Approve (L1 logistic @ tuned cut-off). Stage-2 RAG+LLM: Approve with Conditions (golden set required). Dual-judge panel: Approve as evaluation overlay. DPO/RLAIF policy: Challenger — do not promote yet. Overall: Approve with Conditions under SR 11-7 effective challenge.